

UNIT 1 Cells

3D Unit Planning

Lesson 1 The Characteristics of Cells pp. 4–17

Overview

Objective Students observe magnified images of living things in order to support the claim that living things are made up of tiny structures called cells.

SEP Planning and Carrying Out Investigations

DCI LS1.A Structure and Function

CCC Scale, Proportion, and Quantity

Math and English Language Arts standards and features are detailed on lesson planning pages.

Print and **ONLINE** Student Editions

 Explore **ONLINE!**

ENGAGE

Lesson Phenomenon pp. 4–5

Can You Explain It? How do these tiny structures relate to the onion?

EXPLORE/ EXPLAIN

Identifying Cells pp. 6–9

Investigating the Scale of Cells pp. 10–12



Engineer It Define the Problem



Hands-On Lab Observe Cells with a Microscope

Hands-On Lab Worksheet

CCC Handbook

ELA Handbook

Lab Safety Handbook

Math Handbook

SEP Handbook

ELABORATE

Take It Further pp. 13–14
Microscopes over Time

Take It Further

Making a Microscope
Propose Your Own Path

EVALUATE

Lesson Self-Check pp. 15–17

Lesson Quiz

**Make Your Own
Study Guide**



Hands-On Lab Planning

Observe Cells with a Microscope

 45 minutes

 Pairs

Objective Students compare and contrast living and nonliving things at different scales using a microscope.

Materials

- celery stalk
- celery leaf
- cork, thin slice
- eyedropper
- human hair
- light microscope
- microscope slides with coverslips
- salt
- sand
- tissue paper
- water

Advance Preparation

Before the activity, review safety precautions and microscope use with students. Create stations with materials for students. You may choose to prepare slides yourself. If students prepare slides, tell them NOT to add water to the salt slide.

Setup and Procedure

- During the activity, remind students to hold the slides carefully at the sides so they do not leave fingerprints on the slides that may interfere with their observations.
- Encourage students to observe the materials in the order listed in the table and to repeat Steps 1–6 for each of these materials.
- Remind students to exercise caution when working with glass and to notify you immediately if a piece of glass breaks.

Collaboration

Group Activity Have groups write a comparison of the living and nonliving things that they observed under the microscope. Encourage them to draw pictures of their observations to illustrate their writing. Tell them to write and draw to explain what they observed for someone who does not have access to a microscope.

Lesson 2 Cell Structures and Function pp. 18–35

Overview

Objective Students describe the functions of cells as a whole and the ways in which cell parts contribute to cell function. Students compare these functions to how a sports stadium operates.

SEP Developing and Using Models

DCI LS1.A Structure and Function

CCC Structure and Function

Math and **English Language Arts** standards and features are detailed on lesson planning pages.

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Lesson Phenomenon pp. 18–19

Can You Explain It? How is a cell like a sports stadium?

EXPLORE/ EXPLAIN

Comparing Cell Structures pp. 20–24

Using Cell Models pp. 25–28

 **Hands-On Lab** Use Cell Models to Investigate Cell Size

Explaining Limits to Cell Size pp. 29–30

 **Engineer It** Identify Solutions

Hands-On Lab Worksheet
CCC Handbook
ELA Handbook
Lab Safety Handbook
Math Handbook
SEP Handbook

ELABORATE

Take It Further pp. 31–32
People in Science: Biologist

Take It Further
Understanding Cell
Metabolism
Propose Your Own Path

EVALUATE

Lesson Self-Check pp. 33–35

Lesson Quiz
Make Your Own
Study Guide



Hands-On Lab Planning

Use Cell Models to Investigate Cell Size

 20 minutes

 Pairs

Objective Students use gelatin cubes to model a cell's capacity to move materials across the cell membrane.

Materials

- beaker, 250 mL (2)
- calculator (if desired)
- container, plastic, 473 mL (2)
- gelatin cubes, prepared (1 large and 27 small)
- ruler, metric
- stopwatch or clock with second hand
- water, warm

Advance Preparation

Before the activity, prepare the gelatin using food coloring to make it easier to observe when the cube has dissolved completely in the warm water. Provide goggles, aprons, and gloves for student protection and towels for cleaning up spills.

Setup and Procedure

- During the activity, have students place the two plastic containers of gelatin cubes side by side. This will help them observe and time how long the cubes take to dissolve.
- Students may use phones or other devices to take photographs of the containers while the cubes dissolve.
- When the lab is complete, ask students to pour the contents of the containers carefully down the drain to avoid spills. Have them wipe up any spills to avoid risk of slips and falls.

Differentiated Instruction

RTI/Extra Support Pair students who struggle with math with students who feel more confident with the calculations. Demonstrate how to measure the cubes and use these measurements to replace the variables in the given formulas. Allow students to use calculators to do the math.

Preassessment

The Unit Pretest focuses on prerequisite knowledge and is composed of items that evaluate student's preparedness for the content covered within this unit.

Interactive Worktext, Explorations, Lesson Self-Check

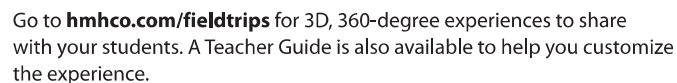
The Lesson Quiz provides a quick assessment of each lesson objective and of the portion of the Performance Expectation aligned to the lesson.

The Performance Task provides students with an opportunity to collaborate with classmates in order to practice or be assessed on aspects of the Performance Expectations aligned to the unit.

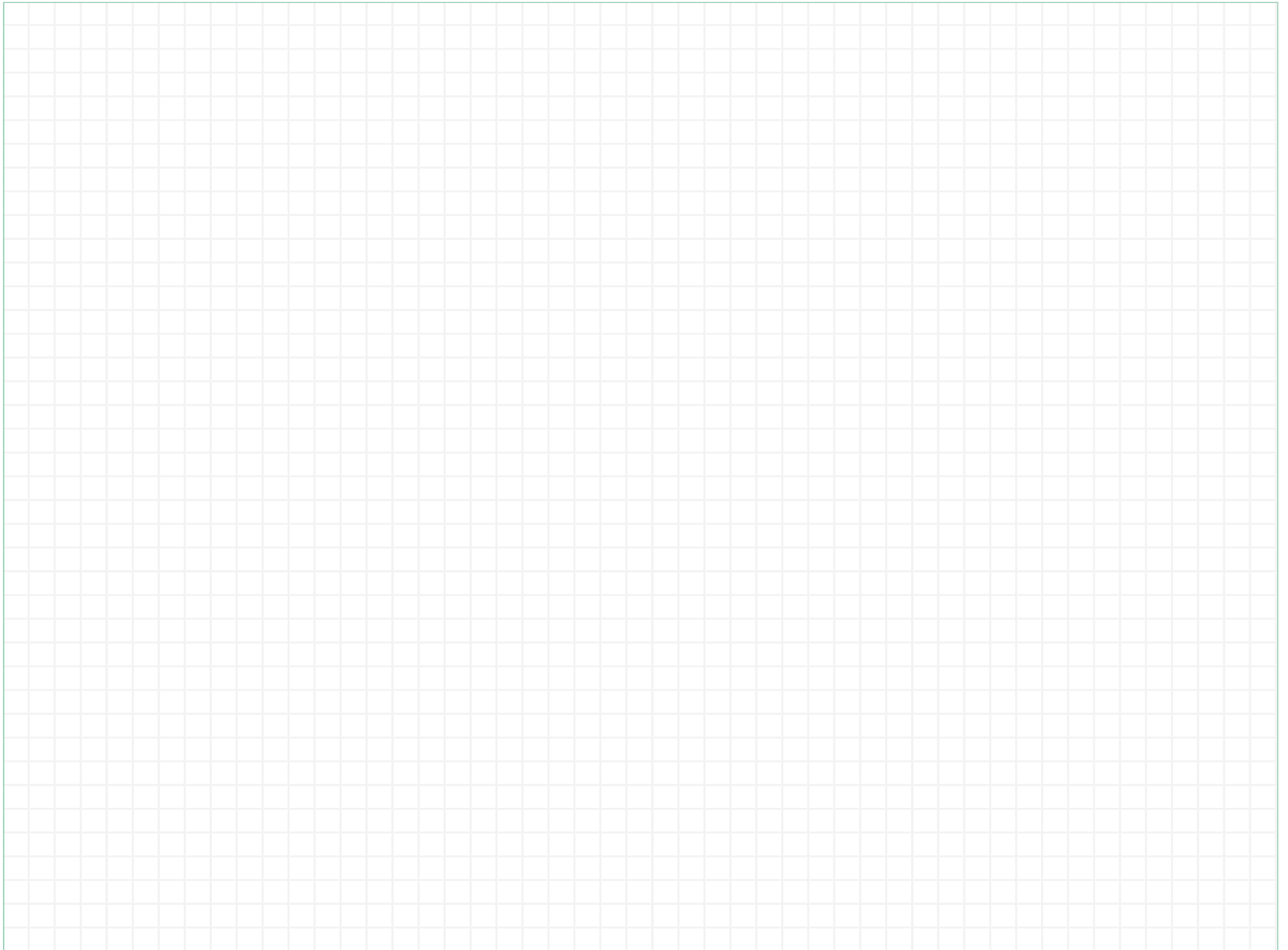
Unit Test A provides an in-depth assessment of the Performance Expectations aligned to the unit. Unit Test B can be used to assess students who need extra support.



Google Expeditions



Teacher Notes



UNIT 1 Cells

Differentiate Instruction

Vocabulary Support

Unit Vocabulary

organism [Lesson 1](#)

cell [Lesson 1](#)

unicellular organism [Lesson 1](#)

multicellular organism [Lesson 1](#)

cell membrane [Lesson 2](#)

organelle [Lesson 2](#)

nucleus [Lesson 2](#)

mitochondria [Lesson 2](#)

chloroplasts [Lesson 2](#)

cell wall [Lesson 2](#)

REINFORCING VOCABULARY To help students remember each vocabulary word introduced in this unit, have them make a Description Wheel for each term as they read it in the text. A Description Wheel has the definition of the vocabulary word and several details surrounding it, fanning out in a circle.

ACADEMIC VOCABULARY The words *process* and *transport* are commonly used in both science and non-science disciplines. A process is a series of actions that produce results. This word can be thought of in a more general sense, but also more specifically in relation to how organisms carry out actions. The word *transport* is often used to describe moving people or items in vehicles, cars, buses, and planes. However, cells can also transport materials and food.

Encourage students to discuss the words *process* and *transport* as they relate to both everyday situations and to science concepts. Have students point out how the words' meanings relate to the context in which they are used.

Differentiate with Technology

ZOOM IN Using a camera or cellular phone, take a photograph of a living organism. Use computer software to magnify a part of the photograph and ask students to describe the details that can be seen on the magnified photograph as compared to the original photograph.

IMAGING OF CELLS Have pairs of students research ways scientists are able to view cells with new state-of-the-art technology, such as 3D imaging. Then have them share their research with the class through a multimedia presentation. Invite discussion about how this new technology may benefit scientists, doctors, and others in understanding diseases.

CHARTING CELL THEORY Ask students to make a digital flow chart of cell theory on a computer. As a class, discuss ways in which digital charts can be used in science. Encourage students to list pros and cons of using digital charts to show science concepts, such as cell theory.

ELL

ELL teaching strategies in this unit:

Lesson 1 p. 4B

Lesson 2 pp. 18B, 22, 25

RTI/Extra Support

Strategies for students who need extra support in this unit:

Lesson 1 pp. 8, 13

Lesson 2 pp. 22, 24, 25, 29, 31

Extension

Strategies for students who have mastered core content in this unit:

Lesson 1 p. 13

Lesson 2 p. 31

UNIT 1 Cells

Making Connections

Connections to Community

Use these opportunities for informal science learning to provide local context and to extend and enhance unit concepts.

At Home

SKIN CELL 3D MODEL Ask students to research what a human skin cell looks like. Have them create a 3D model of a human skin cell with the help of their family members. Students can bring in their models to display in the classroom. *Use with Lesson 2.*

In the Community

MEDICAL INTERVIEW Ask students to interview a doctor, veterinarian, or medical laboratory technician in the community. Have students ask the medical professional how they use information about cells to determine the health of their patients and what differences they notice in healthy and unhealthy cells. Students can present their findings in a short, detailed report. *Use with Lesson 1.*

COMMUNITY ROLES Have students observe the roles of members in the community and document how different people in the community play roles similar to the structures of a cell. For example, police officers and fire fighters are similar to the cell membrane. The cell membrane protects organelles just as police officers and fire fighters protect citizens. Ask students to present their findings. *Use with Lesson 2.*

Culture

BREAD IN CULTURE Bread has a significance in many different cultures. Some types of bread require yeast, a type of unicellular fungus. Other breads do not require yeasts - these types of breads are often called *flat breads*. Have students research a yeast bread and a flat bread from two different regions or cultures. Students can compare and contrast the two types of bread, including the cultural significance, a comparison of the recipes, and an explanation of the role of yeast. *Use with Lesson 1.*



Collaboration

Opportunities for students to work collaboratively in this unit include

Group Discussion pp. 5, 21, 29

Claims, Evidence, and Reasoning p. 8

Feedback pp. 8, 23, 27

Simulation pp. 10, 19

Think, Pair, Share p. 15

Draw, Pair, Share p. 25

Connections to Other Disciplines

Opportunities to connect to other content areas through **Science and Engineering Practices** or **Crosscutting Concepts** in this unit include

Connection to Physical Science pp. 10, 36

Connection to Engineering p. 20

Connection to Art p. 36

Connection to Health p. 36



**Home
Letters**

Use these letters to engage family members with unit concepts.